Investigation of Structural Parameter Effects on Performance of Mach-Zehnder Interferometer

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Abstract: The Mach–Zehnder interferometer (MZI) is a device used to measure relative phase shifts between two beams derived from a single source. In this work, MZIs operating in both TE and TM modes were designed, fabricated, and experimentally characterized. The devices were implemented using strip waveguides with a foundry-standard thickness of 220 nm and a width of 500 nm. Their transmission spectra, effective indices, and group indices were investigated. In addition to MZIs, Y-splitters, microring resonators (MRRs), ring-assisted MZIs (RAMZIs), multimode interferometers (MMIs), 2×2 optical switches, and Bragg gratings were also designed, fabricated, and characterized.

1. Introduction

Over the past decades, integrated photonics has played a significant role in the advancement of optical communications and computing. In particular, silicon photonic devices have demonstrated high-speed integration capabilities, along with improved performance and reduced loss and power consumption. These features have established silicon photonics as a key technology for meeting the growing capacity and bandwidth demands of optical networks.

The MZI is one of the fundamental building blocks in silicon photonic integrated circuits (PICs). It has attracted significant attention and has been widely used in applications such as optical modulation, switching, and routing. In this report, multiple MZI structures are investigated. A conventional MZI, based on strip waveguides with a cross-section of 500 × 220 nm², is analyzed for both the fundamental quasi-TE and quasi-TM modes. The same simulations are then extended to MZI structures with different path lengths.

Figure 1 illustrates the cross-section of a SOI strip waveguide, which consists of 4 layers: the silicon substrate, buried oxide (BOX), the silicon waveguide core, and the oxide cladding. The strip waveguide configuration is ideal for routing purposes and tight bends.

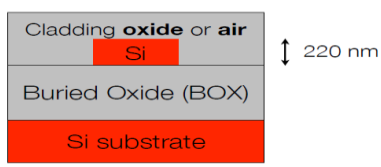


Fig. 1. Cross-section of a strip waveguide

This report is organized as follows: Section 2 briefly introduces the interferometric principles governing the MZI and MRR transfer functions. Section 3 presents the simulation results for the waveguides, Y-splitters, and bends using Lumerical MODE and INTERCONNECT. Section 4 describes the layout masks used for fabrication. Finally, Section 5 analyzes the measurement results of all fabricated devices.

2. Theory

This section presents the theoretical background of the MZI and MRR.

2.1 Mach Zehnder Interferometer (MZI)

A balanced MZI is a 2-port device, formed by a beam-splitter at the input side and a beam-combiner at the output side, connected by waveguides with identical geometries and material properties. With this configuration, for any wavelength, the signals of the two arms of the MZI are constructively combined, in the MZI output.

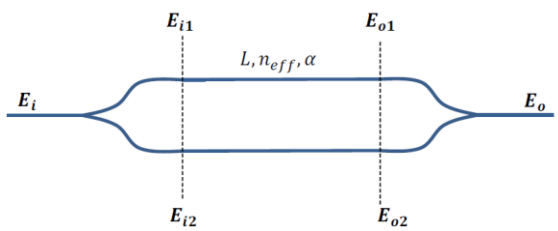


Fig. 2. Balanced MZI: arms have the same length L, effective index n_{eff}, and propagation loss α.

To exploit the different levels of interference, a length asymmetry can be included between the arms, as it is shown in Figure 3. Besides, the two waveguides can present differences in their effective indexes (n_{eff1}, n_{eff2}) and attenuation constants (α_1, α_2). Thus, depending on the signal wavelength, the phase-mismatch between the two combined signals (E_{o1} *and* E_{o2}) causes different kinds of interference, from constructive to destructive.

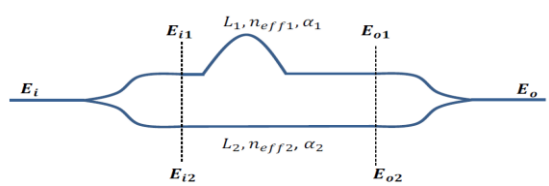


Fig. 3. Unbalanced MZI: Upper and lower arms have distinct properties.

The electrical field at the input of the MZI is represented by E_i. After the Y-splitter, the complex electrical fields at the upper and lower arms are given by:

$$E_{i1} = E_{i2} = \frac{E_i}{\sqrt{2}}$$

The electrical fields at the input of the Y-combiner can be described by:

$$E_{o1} = E_{i1}.e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} = \frac{E_i}{\sqrt{2}}.e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1}$$

$$E_{o2} = E_{i2}.e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} = \frac{E_i}{\sqrt{2}}.e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2}$$

Where $\beta_{1,2} = (2\pi/\lambda)\, n_{eff1,2}$ accounts for the propagation constants of the upper/lower arms. For the Y-combiner, each input field has the amplitude equally divided between the fundamental mode and the radiation modes. Thus,

$$E_o = \frac{1}{\sqrt{2}}(E_{o1} + E_{o2}) = \frac{E_i}{2}\left(e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2}\right)$$

As the light intensity is proportional to the squared power of the field amplitude, the light intensity in the output of the MZI is:

$$I_o = \frac{1}{4} I_i \left| e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right|^2$$

The above equation represents the general form of an unbalanced MZI transfer function and output intensity. However, for the purpose of this work, we can neglect the loss ($\alpha_1 = \alpha_2 = 0$). Besides, the propagation constants may also be considered equal ($\beta_1 = \beta_2 = \beta$) in both arms. Therefore, the equation can be simplified to:

$$I_o = \frac{I_i}{2}\left(1 + Cos(\beta . \Delta L)\right)$$

Where $\Delta L = L_1 - L_2$, $\beta = \frac{2\pi n}{\lambda}$. Since β changes as a function of wavelength, we will get sinusoidal variations versus wavelength. The period of the oscillation is dependent on ΔL and on the waveguide properties.

The period between the peaks of the MZI transfer function, *i.e.*, the free spectral range (FSR), is calculated by:

$$FSR\ [nm] = \frac{\lambda^2}{\Delta L\ .\ n_g(\lambda)}\ , \qquad FSR\ [GHz] = \frac{c}{\Delta L\ .\ n_g}$$

Where n_g is the group index, defined by:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda}$$

2.2 Microring Resonator (MRR)

A microring resonator is a compact optical device consisting of a circular waveguide coupled to a straight bus waveguide, as shown in Figure 4. Light entering the bus can couple into the ring and circulate multiple times. Resonance occurs when the optical path length of the ring matches an integer multiple of the wavelength, leading to constructive interference. At these resonant wavelengths, light is efficiently coupled into the ring, producing distinct transmission dips or peaks. This property makes MRRs useful for filtering, modulation, and sensing applications.

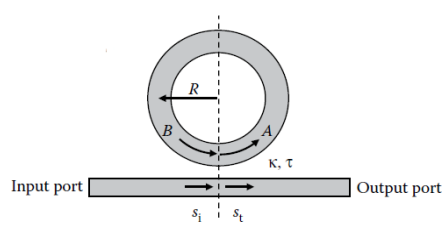


Fig. 4. An all-pass microring resonator (APMR) [8]

The following equation shows the transfer function of an MRR:

$$H_{APMR} = \frac{s_t}{s_i} = \frac{\tau - a_{rt} e^{-j\phi_{rt}}}{1 - \tau a_{rt} e^{-j\phi_{rt}}}$$

Where R represents the radius of the MRR, the coupling coefficient κ describes how much light transfers from the bus waveguide into the ring, and τ is the transmission coefficient representing the portion of light that remains in the bus waveguide. The input optical field entering the bus waveguide is labeled s_i, and the transmitted optical field exiting the bus waveguide is labeled s_t.

2.3 Multi Mode Interferometer (MMI)

A multimode interferometer is a device with many (but a finite number) of guided modes. Each mode has a propagation constant. Because the modes are eigenmodes, they propagate independently from one another. When an MMI is excited by an incident wave, the field profile is decomposed into the eigenmodes. Even though there is no exchange of energy between these eigenmodes, they propagate at a different velocity, and the result is an interference pattern that changes along the length of the MMI.

MMI can be modeled as a device with N-inputs and M-outputs. Similarly to the MZI, the MMI can be configured to work as a combiner or splitter in different ways, making it suitable in applications as optical modulators, multiplexers and switches. MMIs are the most robust for 50/50 splitting due to high fabrication tolerance, while Y-branches offer the broadest bandwidth in the smallest footprint but suffer from scattering at the tip. Directional couplers are the only choice for custom splitting ratios (e.g., 10/90), though they are highly sensitive to wavelength and manufacturing deviations.

As shown in Figure 5, the device is composed of input/output single mode waveguides, input/output tapers and the multimode waveguide. The tapered waveguides were included aiming to achieve an, approximately, adiabatic transition between the single-mode and multimode waveguides. The central waveguide is designed to support a large number of modes. Therefore, the signal propagation in the input/multimode waveguides interface suffers an abrupt transition, exciting multiple modes of the central waveguide. In this region, the propagation is described by the self imaging principle. For a 2×2 MMI, the device length is given by [9]:

$$L_{MMI} = \frac{3\pi}{2(\beta_0 - \beta_1)}$$

where β_0 and β_1 are the propagation constants of the fundamental and first-order modes of the multimode section.

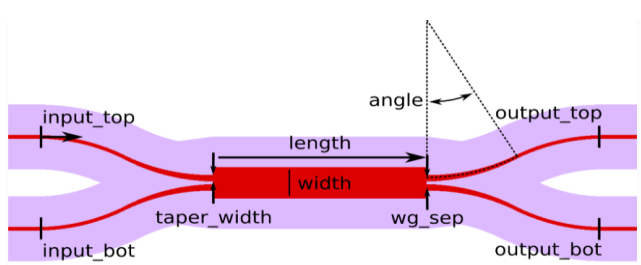


Fig. 5. Schematic of a 2x2 MMI with tapers.

3. Modeling and Simulation

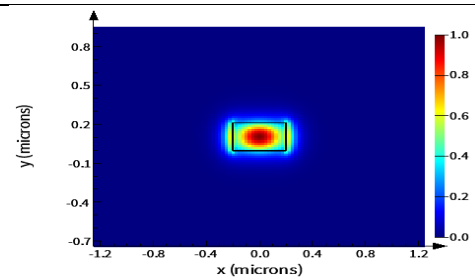
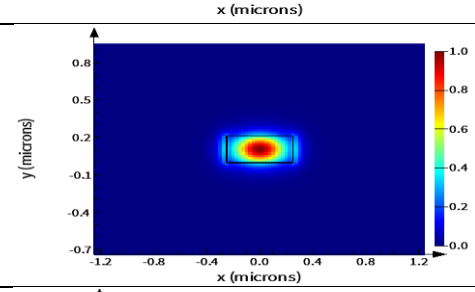
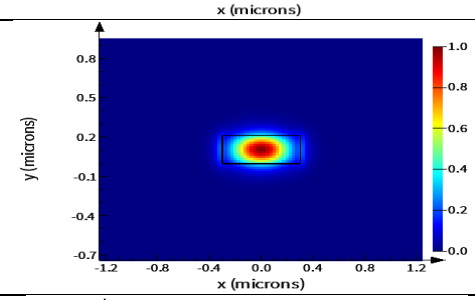
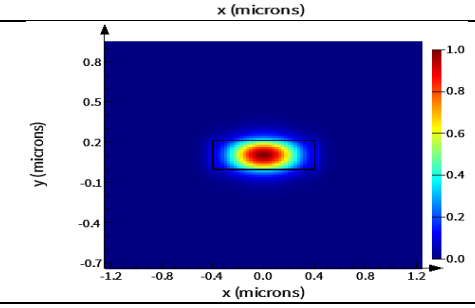
Lumerical MODE and INTERCONNECT were used for mode analysis and circuit simulation, respectively.

3.1 Waveguide Modelling in Lumerical Mode

MODE simulation is done to find the spatial field distribution of TE and TM modes for different waveguide cross-sections. Mode profiles for four cross-sections (400×220, 500×220, 600×220 and 800×220 nm^2) at 1.5 µm wavelength are shown in Tables 1 and 2 for TE and TM modes, respectively.

As the waveguide width increases (from 400 nm to 800 nm), the fundamental TE-mode becomes more tightly confined within the silicon core. This is visible in the heat maps, where the mode "spot" stays centered and fits more comfortably within the larger geometric boundaries. Besides, Table 1 shows that the effective index of the fundamental mode (Mode #1) increases with width.

Table 1. TE-mode profiles for four different cross-sections at 1.5 µm.

	Fundamental TE-mode	Index
400 × 220		
500 × 220		
600 × 220		
800 × 220		

^ mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ex)
1	2.298427+1.319796e-09i	1.5	0.00048019	4.402917+3.029140e-09i	97
2	1.749900+8.610531e-10i	1.5	0.00031328	3.728516+4.948361e-09i	5
3	1.436169+2.973687e-10i	1.5	0.00010819	1.851332+8.409343e-10i	51

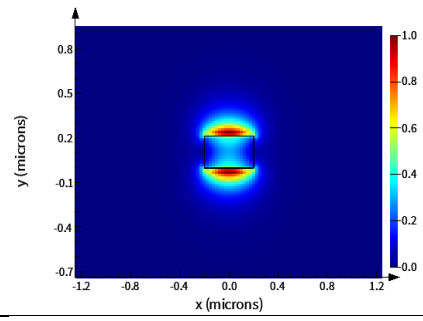
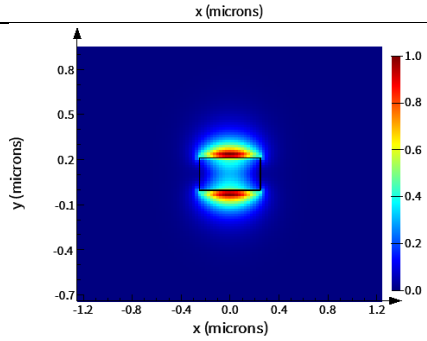
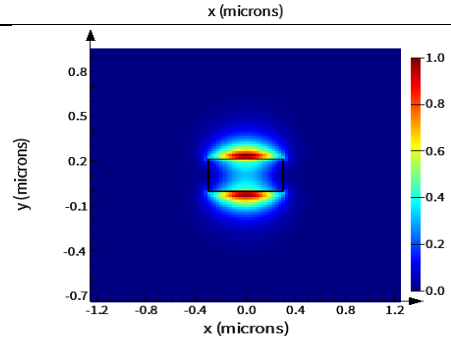
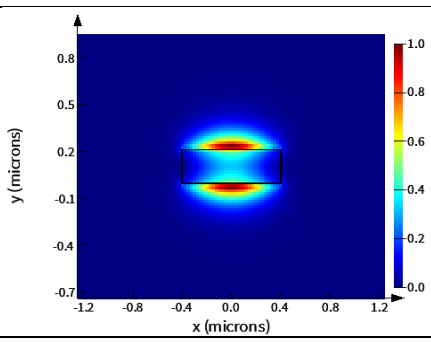
^ mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ex)
1	2.503380+1.297784e-09i	1.5	0.00047218	4.196678+2.802685e-09i	99
2	1.836526+9.718270e-10i	1.5	0.00035358	4.007656+4.785887e-09i	4
3	1.532900+6.249787e-10i	1.5	0.00022739	2.815674+5.056740e-09i	70

^ mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ex)
1	2.616741+1.271579e-09i	1.5	0.00046264	4.061721+2.763219e-09i	99
2	1.899031+1.027928e-09i	1.5	0.00037399	4.129334+4.545809e-09i	4
3	1.778996+1.259895e-09i	1.5	0.00045839	4.511571+6.213950e-09i	88

^ mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ex)
1	2.730194+1.238813e-09i	1.5	0.00045072	3.921045+2.752779e-09i	100
2	2.249621+1.409665e-09i	1.5	0.00051288	4.588104+2.867610e-09i	98
3	1.977028+1.072256e-09i	1.5	0.00039012	4.200783+4.217023e-09i	3

Unlike the TE-mode, the Fundamental TM-mode intensity is concentrated at the upper and lower horizontal interfaces of the silicon core. This is characteristic of TM modes in thin SOI waveguides, where the electric field is primarily perpendicular to the substrate. Moreover, the effective index of the fundamental TM mode increases as the waveguide widens.

Table 2. TM-mode profiles for four different cross-sections at 1.5 µm.

	Fundamental TM-mode	Index
400 × 220		
500 × 220		
600 × 220		
800 × 220		

400 × 220

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500 × 220

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3	1.778996+1.259895e-09i	1.5	0.00045839	4.511571+6.213950e-09i	88

800 × 220

^ mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ex)
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2	2.249621+1.409665e-09i	1.5	0.00051288	4.588104+2.867610e-09i	98
3	1.977028+1.072256e-09i	1.5	0.00039012	4.200783+4.217023e-09i	3

The effective index and group index of both TE and TM modes versus wavelength are shown in Tables 3-6 for different waveguide widths (400, 500, 600 and 800 nm). The effective index is decreasing with wavelength, and the group index is increasing with wavelength.

Table 3. Effective index and group index of the fundamental TE and TM modes for 400×220 nm^2

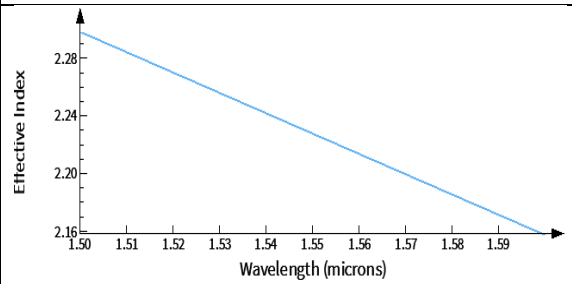
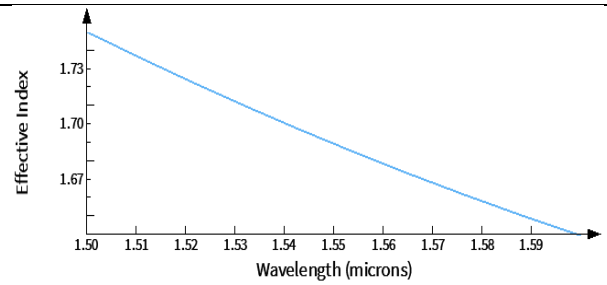
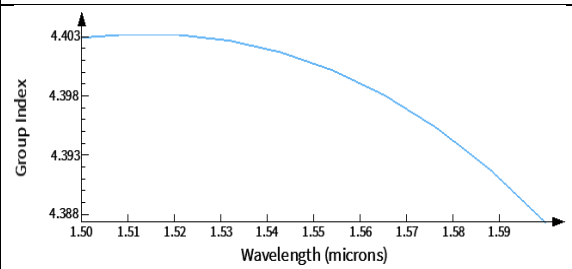
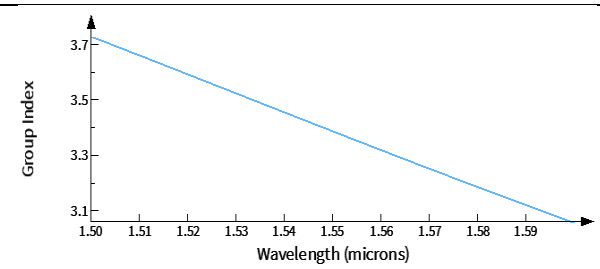
	Fundamental TE-mode	**Fundamental TM-mode**
n_{eff}		
n_g		

Table 4. Effective index and group index of the fundamental TE and TM modes for 500×220 nm^2

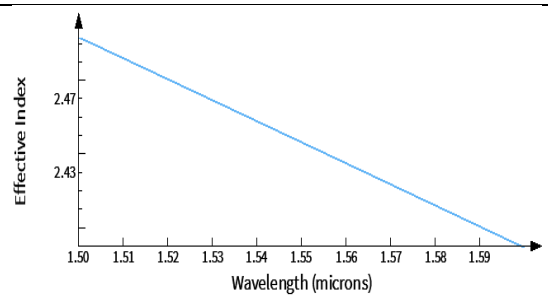
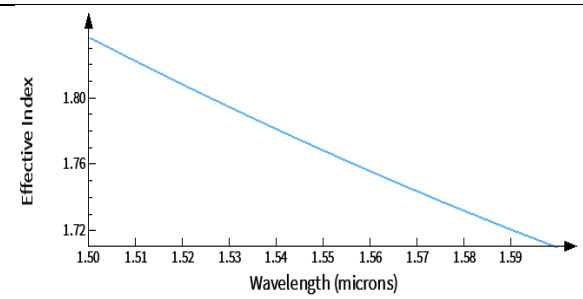
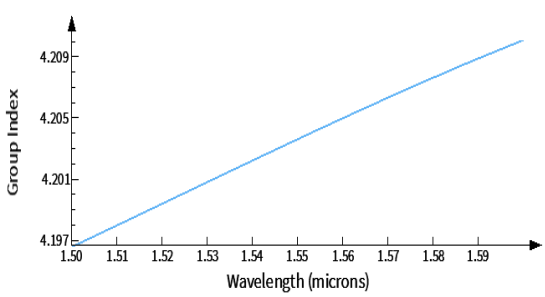
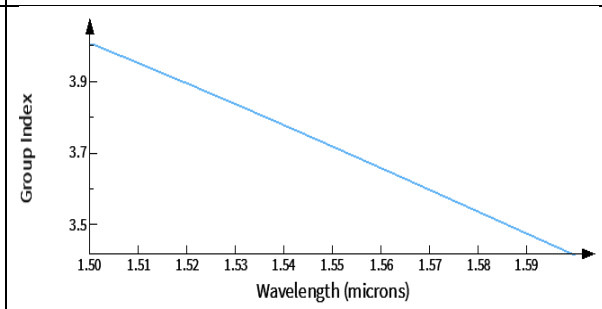
	Fundamental TE-mode	**Fundamental TM-mode**
n_{eff}		
n_g		

Table 5. Effective index and group index of the fundamental TE and TM modes verses for 600×220 nm^2

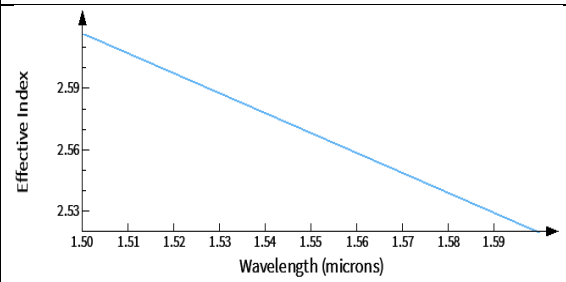
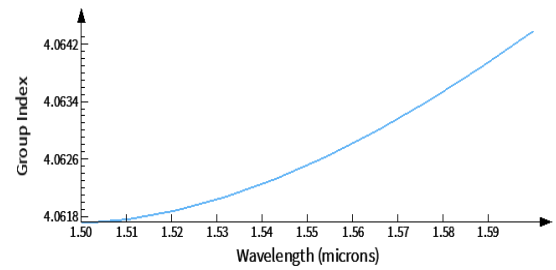
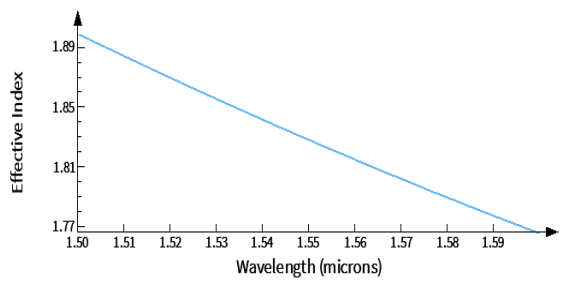
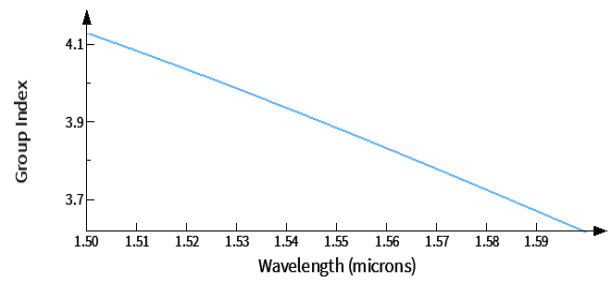
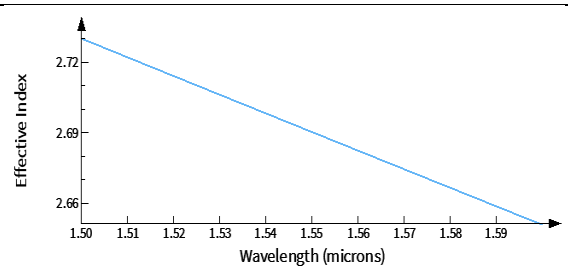
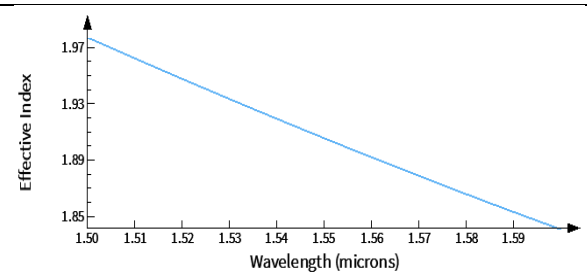
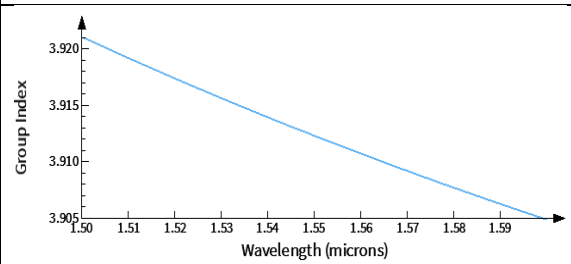
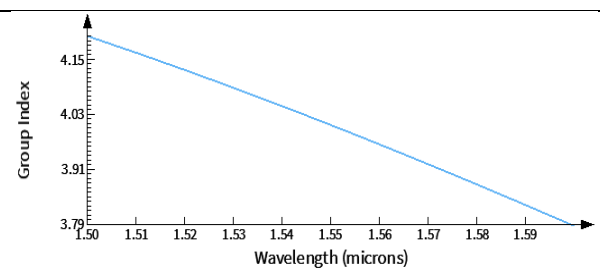
	Fundamental TE-mode	Fundamental TM-mode
n_{eff}		
n_g		

Table 6. Effective index and group index of the fundamental TE and TM modes for 800×220 nm^2

	Fundamental TE-mode	Fundamental TM-mode
n_{eff}		
n_g		

3.2 Waveguide Compact Model

Using the data exported from frequency sweep, the three coefficients of the Taylor expansion for compact model are found for all waveguide geometries and for both polarizations using a script in Lumerical MODE. The Taylor expansion is:

$$n_{eff,}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

The Taylor expansion parameters can also be converted to conventional parameters (effective index, group index, and dispersion) at the wavelength λ_0:

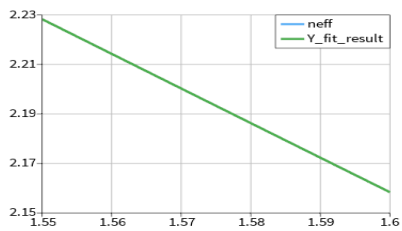
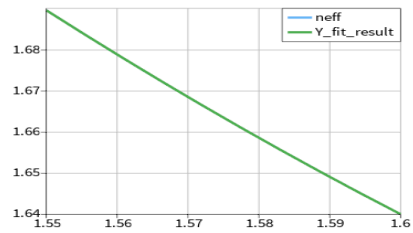
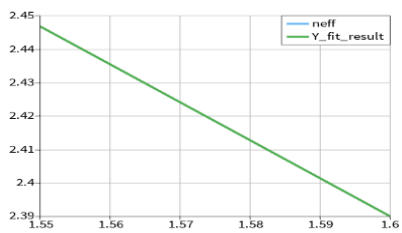
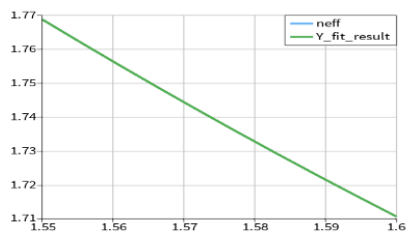
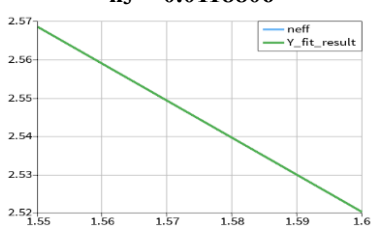
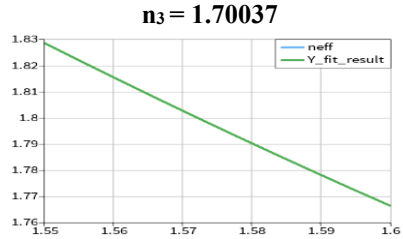
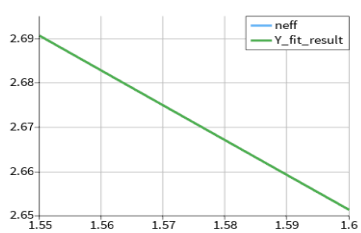
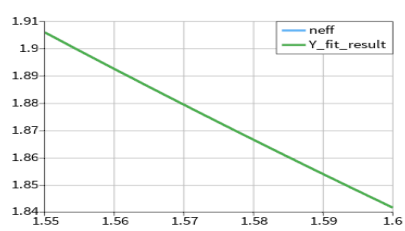
$$n_{eff} = n_1$$

$$n_g = n_1 - n_2\,\lambda_0$$

$$D = -2\,\lambda_0 \frac{n_3}{c} \quad [s/m^2]$$

The coefficients n_1, n_2, and n_3 are listed in Table 7 for all waveguides and polarizations.

Table 7. Taylor expansion coefficients for different waveguides (λ_0=1.55 µm)

Cross section	Fundamental quasi-TE	Fundamental quasi-TM
400×220	**n_1 = 2.22829, n_2= -1.40236** **n_3 = 0.0845582** 	**n_1 = 1.68953, n_2 = -1.09461** **n_3 = 2.08537** 
500×220	**n_1 = 2.44682, n_2 = -1.13352** **n_3 = -0.041065** 	**n_1 = 1.76882, n_2 = -1.25896** **n_3 = 1.92378** 
600×220	**n_1 = 2.56856, n_2 = -0.963796** **n_3= -0.0118806** 	**n_1 = 1.82861, n_2 = -1.3282** **n_3 = 1.70037** 
800×220	**n_1 = 2.69065, n_2 = -0.788107** **n_3 = 0.0468452** 	**n_1 = 1.906, n_2 = -1.35716** **n_3 = 1.38996** 

3.3 Waveguide Temperature Dependence

We extracted the temperature dependence of an optical waveguide. The silicon thermo-optic coefficient is $\frac{\mathbf{dn}}{\mathbf{dT}} = \mathbf{1.86 \times 10^{-4}\ K^{-1}}$. We wish to find the change in the waveguide effective index versus temperature $\frac{dn_{eff}}{dT}$. We wanted to see the effect of 10 degrees change in the temperature in the effective index of a 500×220 nm^2 waveguide. The thermo-optic coefficient for a 500×220 nm^2 waveguide is **0.000198528**.

```
> ?neff_0=abs(getdata('FDE::data::mode1','neff'));
result:
2.44095
> ?neff_10=abs(getdata('FDE::data::mode1','neff'));
result:
2.44293
> ?(neff_10 - neff_0)/10;
result:
0.000198528
```

3.4 Y-Branch Simulations using Lumerical MODE 2.5D FDTD

We imported the GDS file of the designed Y-branch to Lumerical. Then we added a varFDTD solver, a source, and a monitor. Figure 6 shows the Y-branch simulated in Lumerical MODE.

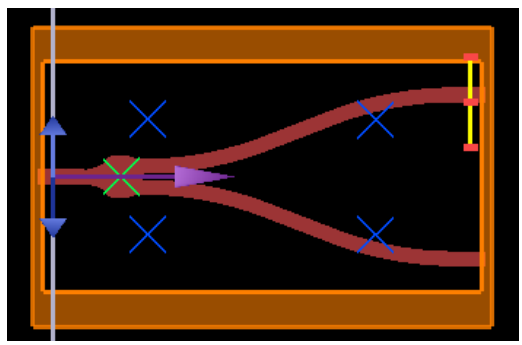


Fig. 6: Y-branch in Lumerical MODE

Figure 7 shows the transmission (T) of the Y-branch versus wavelength that shows approximately 50:50 split ratio for the two outputs of the Y-branch.

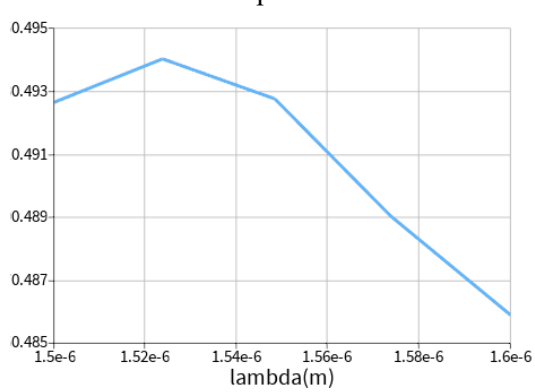
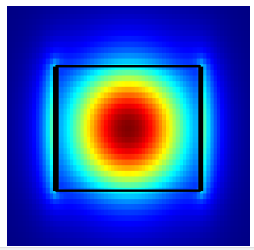


Figure 7: Y-branch transmission spectrum

3.5 Waveguide Bends Simulation using Lumerical MODE

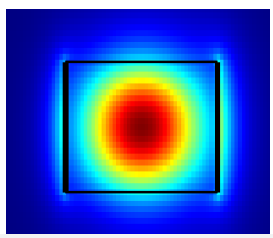
Light propagating in a bend gets pushed out to the side. This will lead to two additional types of losses, **mode mismatch losses** and **radiation losses**. Mode mismatch is the dominant source of loss in TE bends. Radiation loss is very small in strip waveguides. However, it does play a role in rib waveguides.



mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ey)	waveguide TE/TM fraction (%)	effective area (µm^2)
1	2.445105+7.825564e-17i	1.55	2.7554e-11	4.243565-1.212078e-17i	98	76.35 / 81.87	0.190022
2	1.779235+3.249014e-12i	1.55	1.1440e-06	3.670297-3.150799e-10i	4	71.1 / 88.95	0.359715
3	1.491773-2.572754e-09i	1.55	-0.00090586	2.507514+1.015294e-07i	57	86.34 / 95.91	0.726393

Figure 8: mode profile of a **waveguide** in TE mode

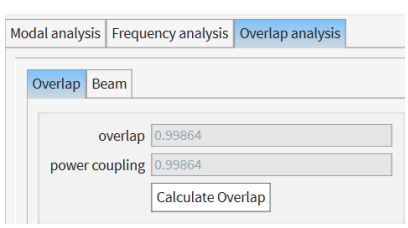
Next, we do the simulation again for a bend radius of **5 µm** in TE mode. We see that the field profile has shifted to the right, and the optical loss has also increased.



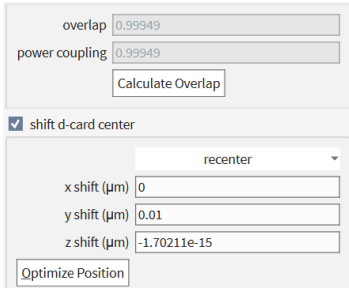
mode #	effective index	wavelength (µm)	loss (dB/cm)	group index	TE polarization fraction (Ey)	waveguide TE/TM fraction (%)	angular propagation constant (1/rad)
1	2.446507-8.422226e-16i	1.55	-2.9654e-10	4.246009+5.161234e-14i	98	76.4 / 81.86	49.58664-1.707045e-14i
2	1.790098+0.0007632261i	1.55	268.73	3.520834-0.06812402i	4	73.66 / 89.54	36.28231+0.01546933i
3	1.519737+0.005764273i	1.55	2029.6	2.514448-0.03877839i	61	82.47 / 95.26	30.80256+0.1168322i
4	1.182686+0.01492157i	1.55	5253.8	2.284349+0.03937320i	7	92.6 / 64.17	23.97109+0.3024354i
5	1.040955+0.02252275i	1.55	7930.2	2.416796+0.2322558i	10	91.68 / 53.98	21.09844+0.4564988i
6	0.9849098+0.01430219i	1.55	5035.8	2.859993-0.4700329i	11	66.2 / 75.38	19.96249+0.2898817i

Figure 9: mode profile of a **bend** in TE mode

By doing the overlap analysis, we calculate the power coupling between the straight and the bent waveguides. We see that there's a 0.2% loss.



The optimization finds that a shift of 10 nm results in a reduced loss of 0.085%, nearly a factor of 2.35 improvement.



We can repeat the same procedure to find the loss for the **TM** mode. This time we selected the second mode.

mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	TE polarization fraction (Ey)	waveguide TE/TM fraction (%)	effective area (μm^2)
1	2.445105+7.352034e-18i	1.55	2.5886e-12	4.243565-6.856143e-17i	98	76.35 / 81.87	0.190022
2	1.779235+3.249011e-12i	1.55	1.1440e-06	3.670297-3.150793e-10i	4	71.1 / 88.95	0.359715
3	1.491773-2.572754e-09i	1.55	-0.00090586	2.507514+1.015295e-07i	57	86.34 / 95.91	0.726393

For a 5 μm bend radius, we find that the mode is much more strongly affected by the bend, and the coupling loss is 7%, much higher than the TE case.

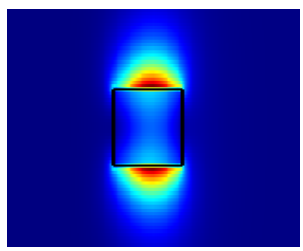
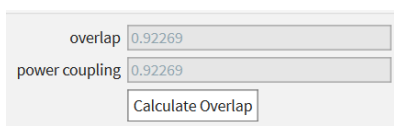
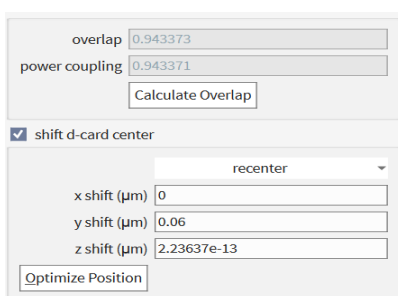


Figure 10: mode profile of a waveguide in TM mode



Optimizing the position shows that a 60 nm offset reduces the loss to 5%.



3.6 Lumerical Interconnect

We provided the S-parameter file to the INTERCONNECT and plotted the insertion loss of the Y-branch as shown in Figure 11. The insertion loss is higher than 3 dB as expected for a real physical component.

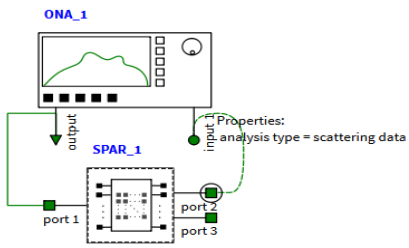


Figure 11: Y-branch in Interconnect

Figure 12: IL vs. wavelength for the Y-branch

Next, we characterize the MZI using 2 Y-branches and 2 waveguides. Figure 13 shows the MZI circuit in INTERCONNECT. Note that there is a 100 µm length difference between the two arms of the MZI.

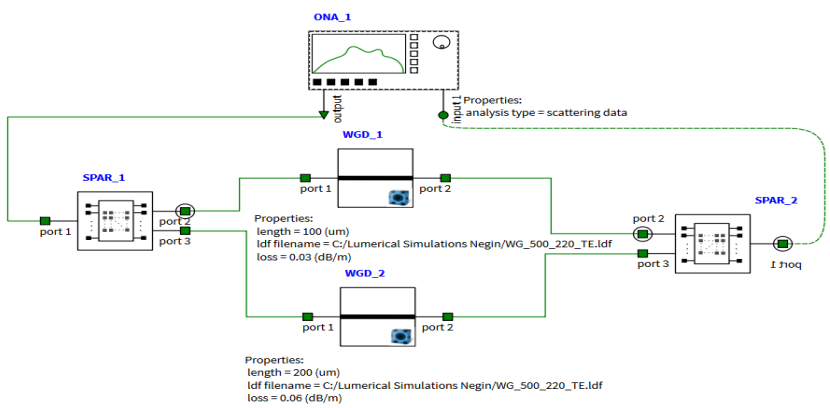


Figure 13: MZI in INTERCONNECT

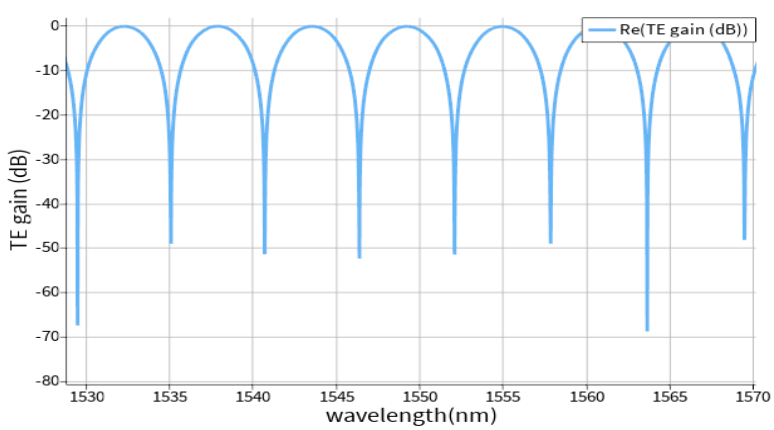


Figure 14: MZI TE gain in dB vs. wavelength

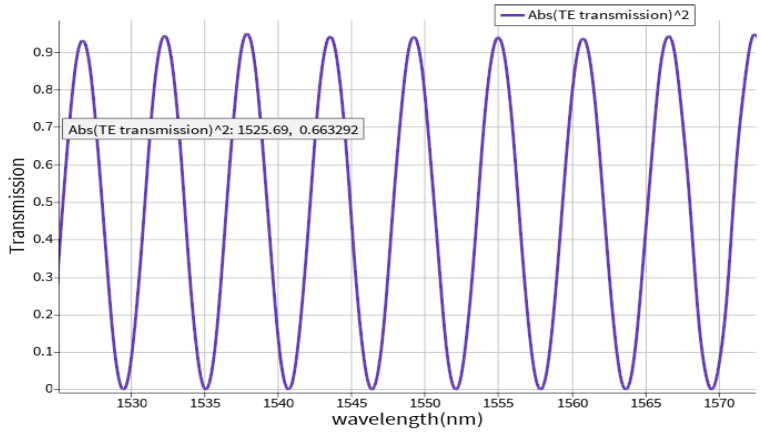


Figure 15: MZI TE transmission vs. wavelength

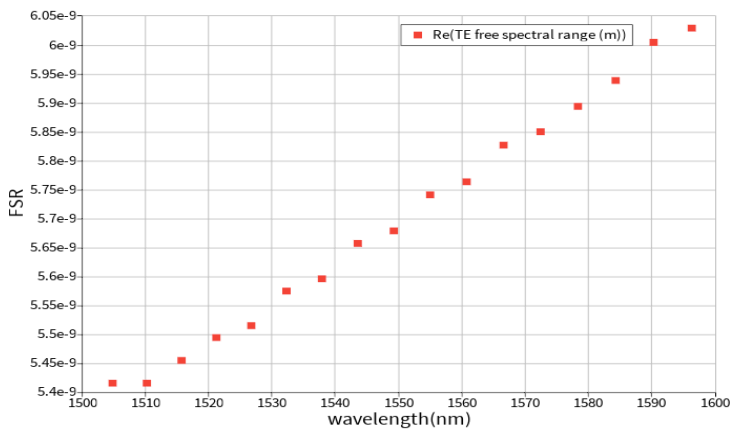


Figure 16: MZI FSR vs. wavelength

A script prompt was used to find the FSR:

```
> ?1550e-9^2/100e-6/4.2;
result:
5.72024e-09
> ?getresultdata('ONA_1', 'input 1/mode 1/peak/free spectral range');
result:
6.02855e-09
6.0041e-09
5.93817e-09
5.89382e-09
5.84997e-09
5.8269e-09
5.76356e-09
5.74115e-09
5.67906e-09
5.65728e-09
5.5964e-09
5.57524e-09
5.51553e-09
5.49496e-09
5.45541e-09
```

Next, we compare the transmission of the two MZI, one with 100 µm and the other with a 200 µm path difference between the arms. The FSR is two times smaller for the 200 µm mismatch as shown in Figure 17.

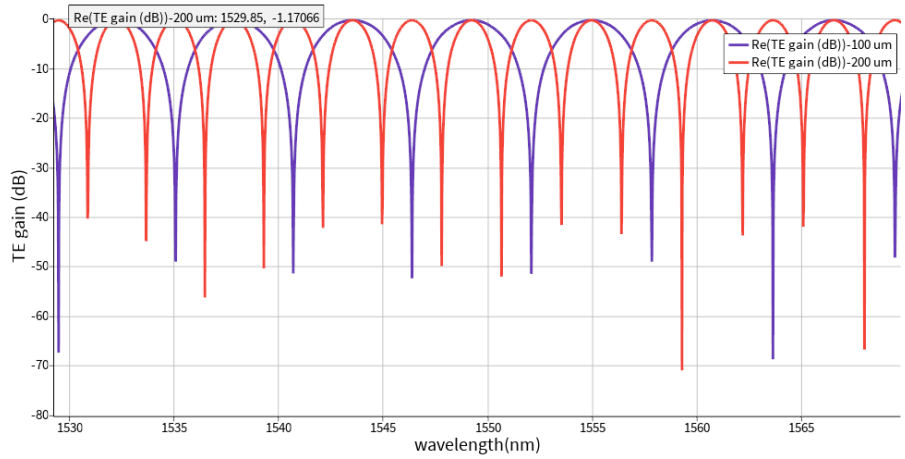


Fig. 17: TE gain in dB vs. wavelength for two MZI with 100 µm and 200 µm mismatch

Next, we add the fibre grating coupler to the MZI circuit. Figure 18 shows the circuit in INTERCONNECT.

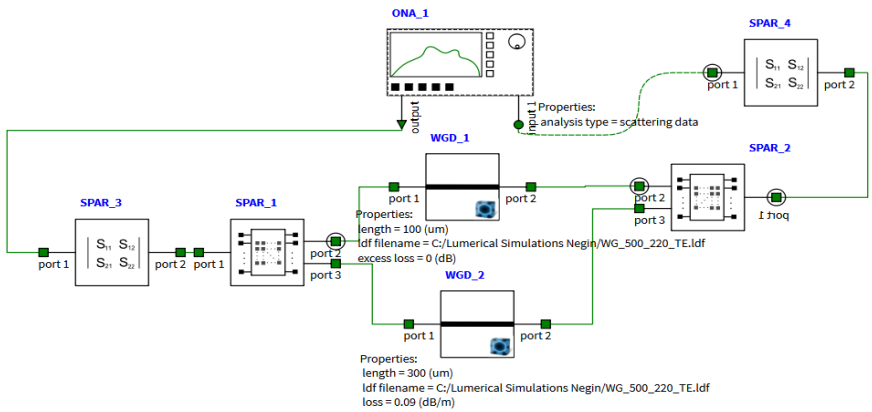


Fig. 18: MZI with grating couplers in INTERCONNECT

At 1550, the insertion of the circuit without grating couplers is about 0.3 dB. Whereas, with the two grating couplers, it is over 5 dB (Figure 19).

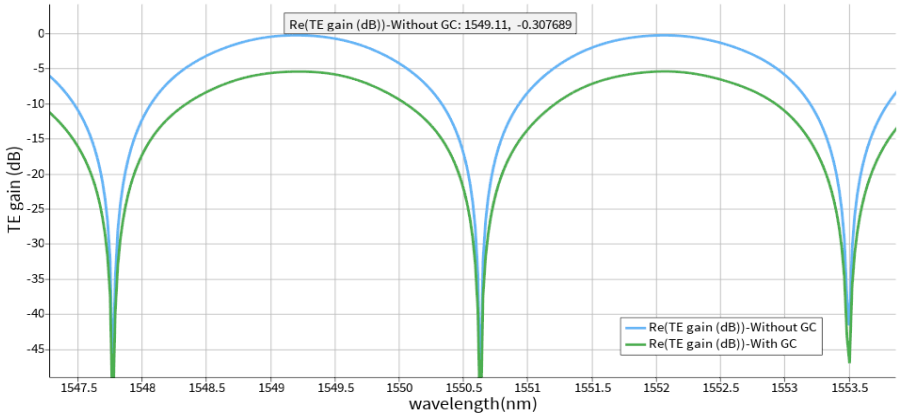


Fig. 19: MZI with and without grating couplers

The best-case insertion loss for the one and two grating couplers was simulated and found to be 2.55 dB and 5.06 dB, respectively. The grating coupler's transfer function, depicted in Figure 21, shows insertion losses ranging from 2.55 dB to 11 dB.

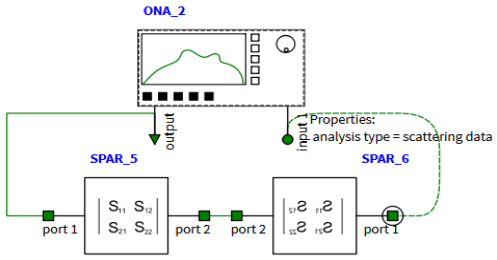


Fig. 20: **Two** grating couplers connected to a network analyzer.

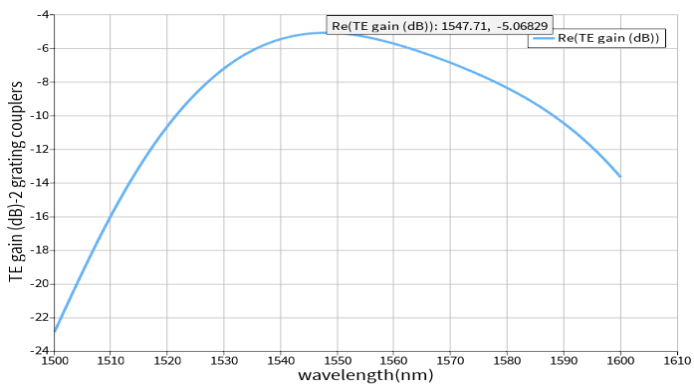


Fig. 21: Transmission of two grating couplers connected to a network analyzer.

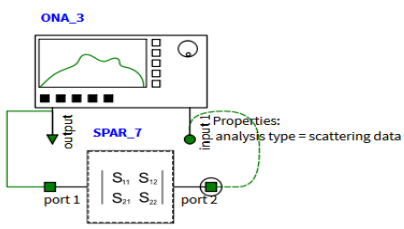


Fig. 22: **One** grating coupler connected to a network analyzer.

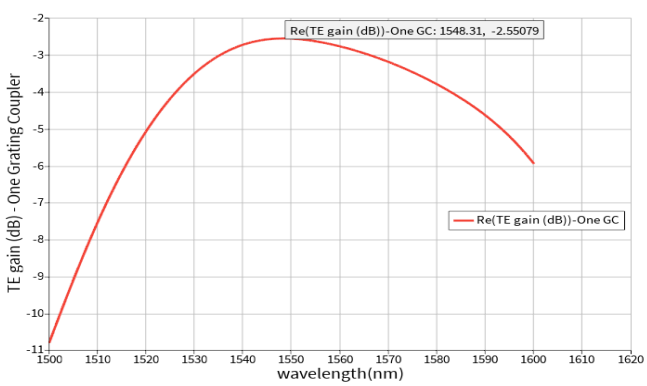


Fig. 23: Transmission of one grating coupler connected to a network analyzer.

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3.7 Building Interferometer circuits using the Compact Model Library

In this section, we simulated a MZI using the Ebeam compact model library. Figures 24 and 25 show the MZI using Y-branches and directional couplers, respectively.

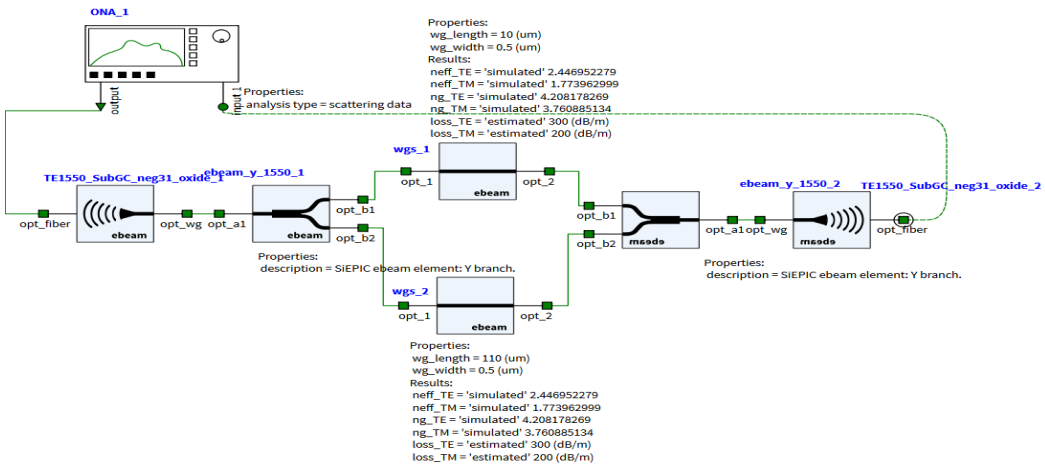

Fig. 24: MZI with Y-branches in INTERCONNECT

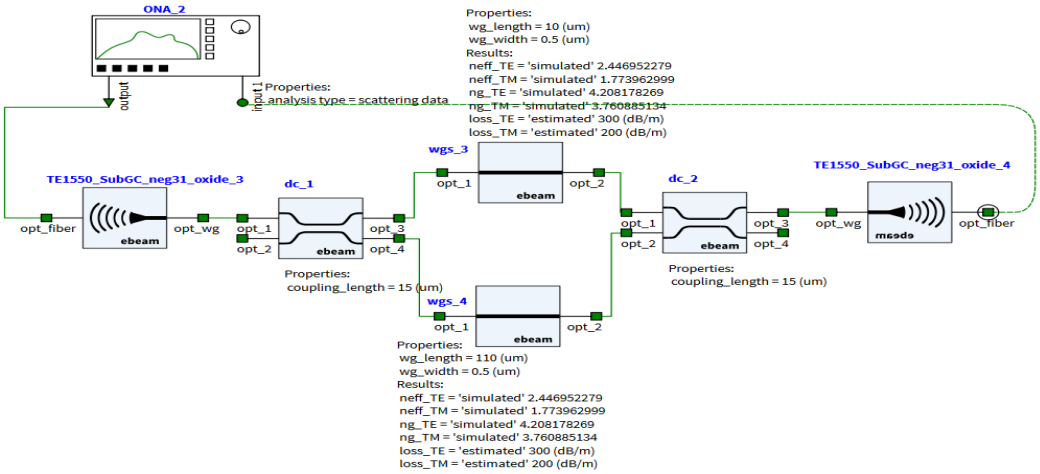

Fig. 25: MZI with directional couplers in INTERCONNECT

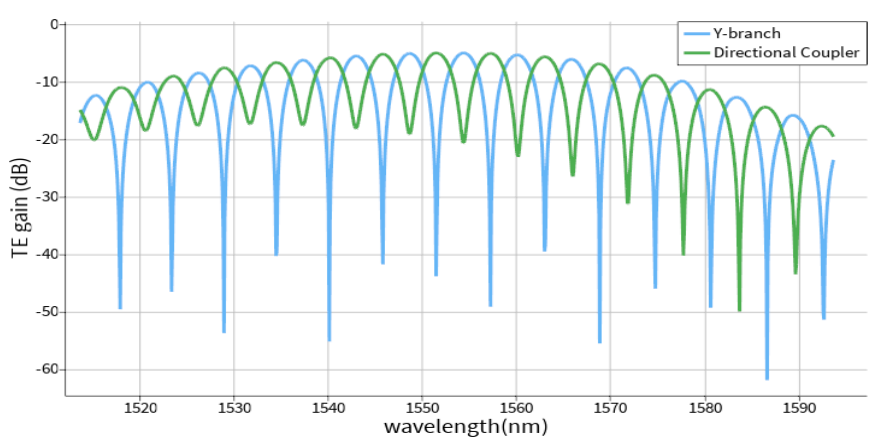

Fig. 26: Transmission of the MZI with Y-branch (blue) and directional coupler (green)

The extinction ratio using the directional couplers is not as large as with the Y-branch. There is a wavelength dependance to the extinction ratio when using directional couplers. You can see for longer wavelengths the extinction ratio is increased (Fig. 28).

Next, we uses adiabatic couplers, and connect the two outputs to the ONA inputs.

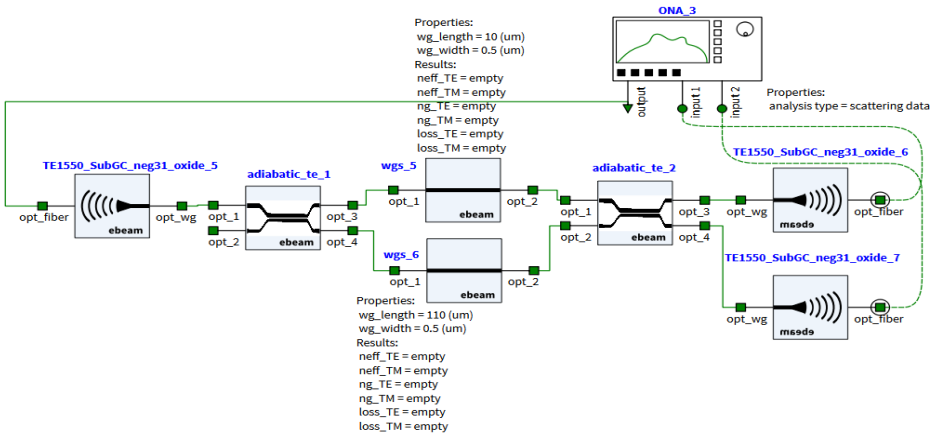

Fig. 27: MZI with adiabatic couplers in INTERCONNECT

The two outputs of this circuit are complementary as you can see in Fig. 27.

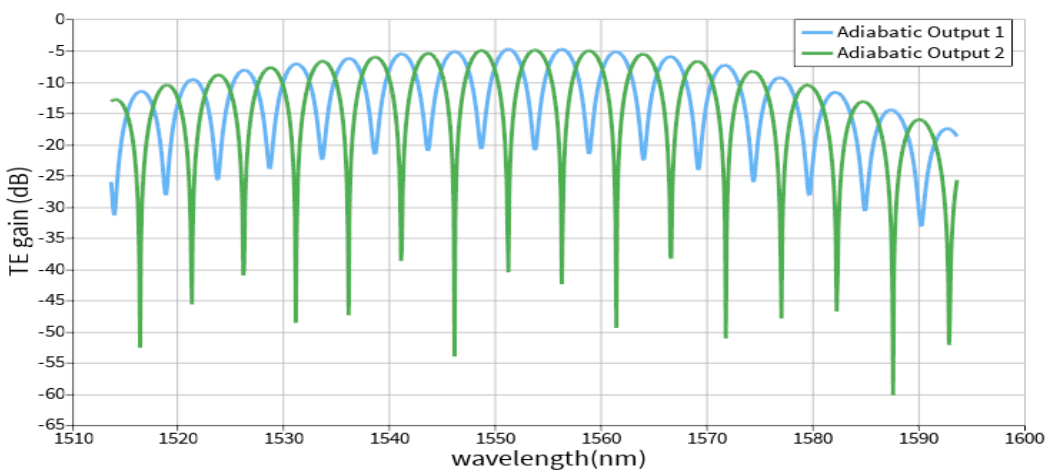

Fig. 28: Transmission of the MZI with adiabatic couplers

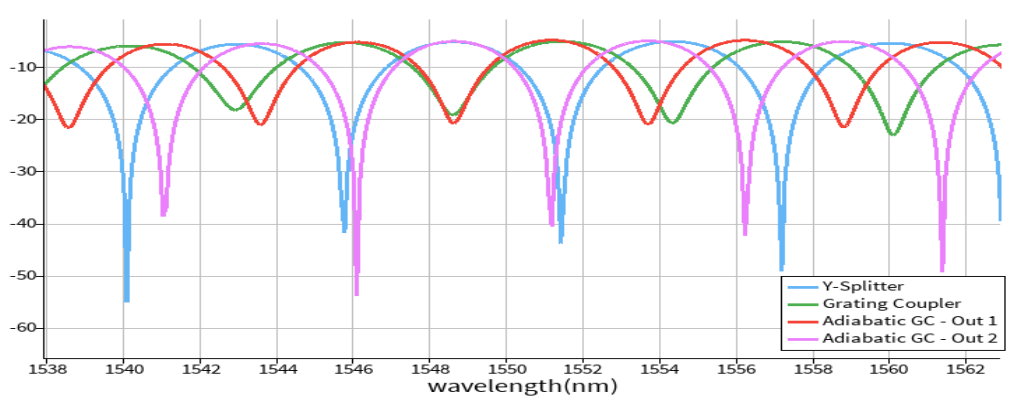

Fig. 29: Comparison of using the Y-splitter, grating coupler, and adiabatic grating coupler in MZI

3.8 Designed Layout Simulation

In this section, we simulated two MZIs and one MRR designed in KLayout for fabrication to extract the free spectral range (FSR) and group index. These values will later be compared with those extracted from measurement results.

3.8.1 TE MZI Simulation

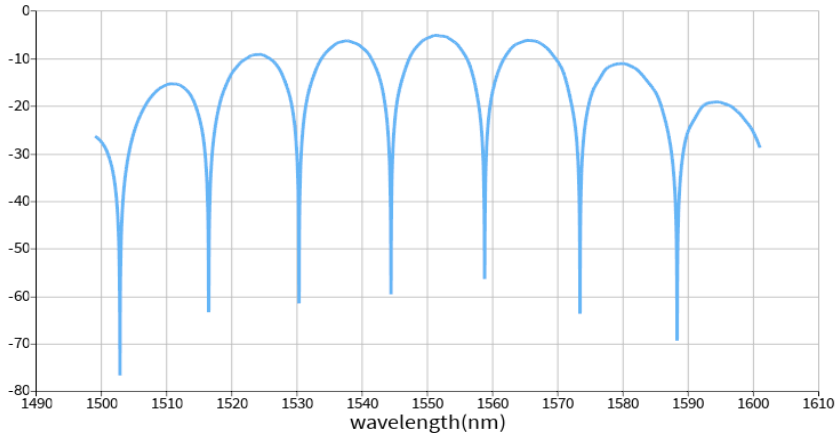

Fig. 30: Simulated transmission spectrum of a **TE** mode MZI with **ΔL= 40 µm**

3.8.2 TM MZI Simulation

- The measurement results for the TM MZI with ΔL= 30.035 µm was **missing**. Therefor, I had to simulate and use the data of the other TM MZI with ΔL= 12.92 µm

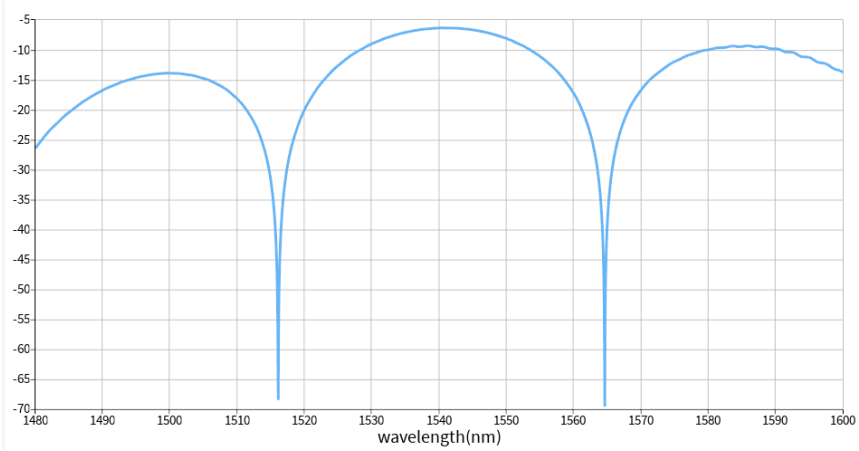

Fig.31: Simulated transmission spectrum of a **TM** mode MZI with **ΔL= 12.92 µm**

3.8.3 TE MRR Simulation

We simulated the microring resonator with the following parameters: R= 10 µm, Gap= 200 nm, Ls=0 µm.

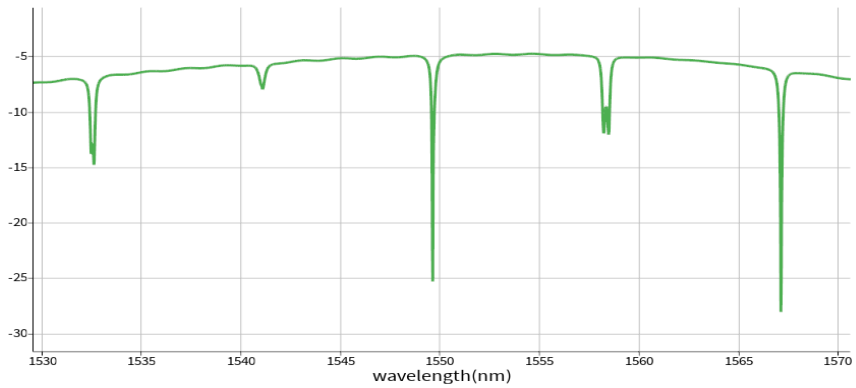

Fig.32: Simulated transmission spectrum of a **TE** mode MRR with R= 10 µm, Gap= 200 nm, Ls=0 µm

Table 8. Simulated values of the FSR and group index for the TE/TM mode MZIs and TE MRR

Waveguide (nm^2)	Mode	ΔL (µm)	FSR (Sim.)	n_g (Sim.)
500 × 220 MZI	TE	40	14.22	4.2238
500 × 220 MZI	TM	12.92	48.49	3.8349
500 × 220 MRR	TE	L=2πR	8.61	4.44

4. Layout

The layout design was created in KLayout, as shown in Figures 33–35. To facilitate future automated testing, the TE and TM grating couplers were oriented to the right, minimizing the need for chip rotations and speeding up measurements. Additionally, a 127 µm pitch was set between the gratings to match the fiber array used in the automated setup. Each layout has a size of 410 µm × 605 µm.

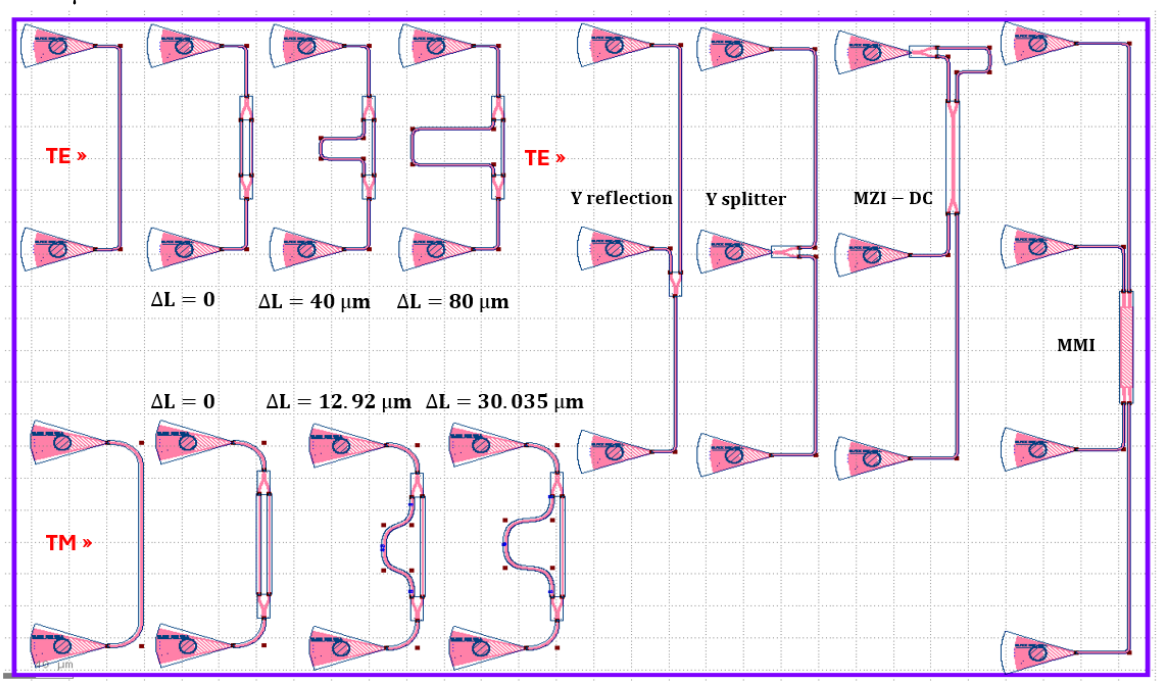

Fig. 33: Layout 1 in Klayout

Table 9 shows the list of devices in Fig. 33:

Table 9. list of devices in layout 1

Layout	Mode	Wavelength	Device	Comment
1	TE	1550	Loop Back	
1	TE	1550	MZI	Balanced
1	TE	1550	MZI	ΔL = 40 µm
1	TE	1550	MZI	ΔL = 80 µm
1	TM	1550	Loop Back	
1	TM	1550	MZI	Balanced
1	TM	1550	MZI	ΔL = 12.92 µm
1	TM	1550	MZI	ΔL = 30.035 µm
1	TE	1550	Y Reflection	
1	TE	1550	Y Splitter	
1	TE	1550	MZI - DC	With directional coupler, ΔL = 41.9 µm
1	TE	1550	2x2 MMI	2 input-2 output

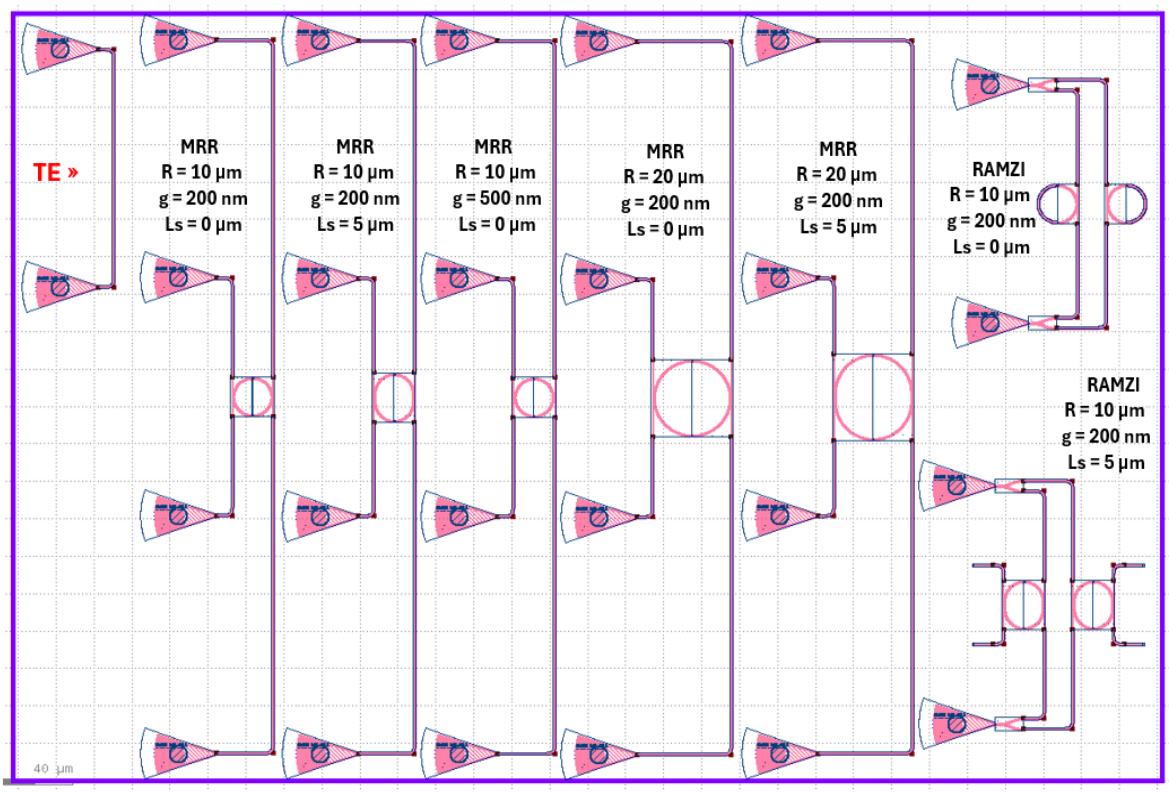

Fig. 34: Layout 2 in Klayout

Table 10 shows the list of devices in Fig. 34. (R: Radius, Ls: Straight length)

Table 10. list of devices in layout 2

Layout	Mode	Wavelength	Device	Comment
2	TE	1550	Loop Back	
2	TE	1550	MRR	R= 10 µm, Gap= 200 nm, Ls=0 µm
2	TE	1550	MRR	R= 10 µm, Gap= 200 nm, Ls= 5 µm
2	TE	1550	MRR	R= 10 µm, Gap= 500 nm, Ls= 0 µm
2	TE	1550	MRR	R= 20 µm, Gap= 200 nm, Ls= 0 µm
2	TE	1550	MRR	R= 20 µm, Gap= 200 nm, Ls= 5 µm
2	TE	1550	RAMZI	R= 10 µm, Gap= 200 nm, Ls= 0 µm
2	TE	1550	RAMZI	R= 10 µm, Gap= 200 nm, Ls= 5 µm

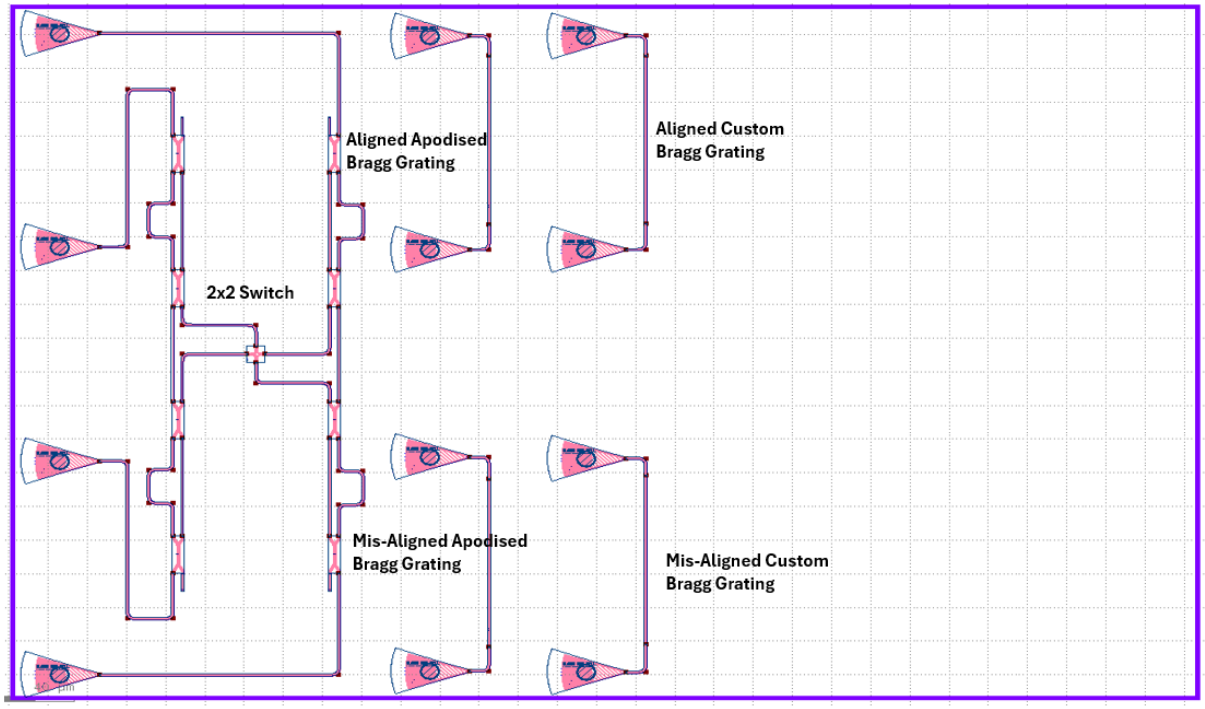

Fig. 35: Layout 3 in Klayout

Table 11 shows the list of devices in Fig. 35:

Table 11. list of devices in layout 3

Layout	Mode	Wavelength	Device
3	TE	1550	2x2 Switch
3	TE	1550	Aligned Apodized Bragg Grating
3	TE	1550	Aligned Custom Bragg Grating
3	TE	1550	Mis-aligned Apodized Bragg Grating
3	TE	1550	Mis-aligned Custom Bragg Grating

5. Fabrication

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm

diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

5.1. Manufacturing Variability

The dominant manufacturing parameter that results in optical component and circuit variation has been identified to be the **top silicon thickness variation.** In this work, the wafers were provided by *Soitec*, Grenoble, France, and they have the following characteristics:

- Wafer diameter: 6";
- Si thickness: average size of 219.2 nm, with a standard deviation of ± 3.9 nm, and a 6 sigma deviation of 23.4 nm;
- SiO2 thickness: average size of 2994.5 nm, with a 6 sigma deviation of 6.3 nm.

The next manufacturing parameter that results in optical component and circuit variation is the **feature size variation**. The variation of feature size is dependent on the fabrication steps. For the E-beam process, some of the chips had features reduced by up to -30 nm, and in some chips increased by up to +10 nm, and the within-chip variations were typically between ± 10 nm. When designing layout, the closer the components, the more correlated the variations are.

5.2. Corner Analysis

We consider process variations in two dimensions, namely the thickness and width variations.
Corner Analysis is a deterministic "worst-case" approach that tests if a design functions at the extreme limits of fabrication (e.g., the thickest/widest and thinnest/narrowest possible waveguides), ensuring the device is robust across the entire process window.

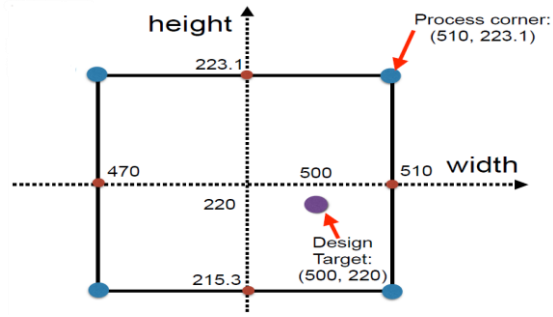

Fig. 36: Corner analysis in photonics

We performed the corner analysis on the 500×220 nm^2 waveguide, for TE and TM modes, at 1550 nm, for the 4 corners, and 1 nominal design. According to the corner analysis, the group index for waveguides is between **4.179** and **4.262** for the **TE mode**, and between **3.480** and **3.836** for the fundamental **TM mode** at 1550 nm.

Table 12. Corner analysis in 1550 nm

Mode	WG Size (nm^2)	Group Index	Effective Index	FSR (ΔL(TE) = 40 µm, ΔL(TM) = 30.035 µm)
TE	510 × 223.1	4.187808	2.472207	14.342 nm
TE	510 × 215.3	4.179682	2.441607	14.370 nm
TE	470 × 223.1	4.262331	2.403880	14.091 nm
TE	470 × 215.3	4.253259	2.372448	14.122 nm
TE	500 × 220	4.203676	2.446821	14.288 nm
TM	510 × 223.1	3.836366	1.796249	20.850 nm
TM	510 × 215.3	3.586497	1.744253	22.303 nm
TM	470 × 223.1	3.743125	1.766349	21.370 nm
TM	470 × 215.3	3.480459	1.716677	22.983 nm
TM	500 × 220	3.720462	1.768817	21.500 nm

5.3. Monte Carlo

Monte Carlo Analysis is a statistical approach that uses random sampling and probability distributions to simulate hundreds of "real-world" variations, allowing you to predict the actual manufacturing **yield** and understand how local mismatches between components will affect system performance. Monte Carlo simulations were performed for the MZI devices shown in green in Table 9.

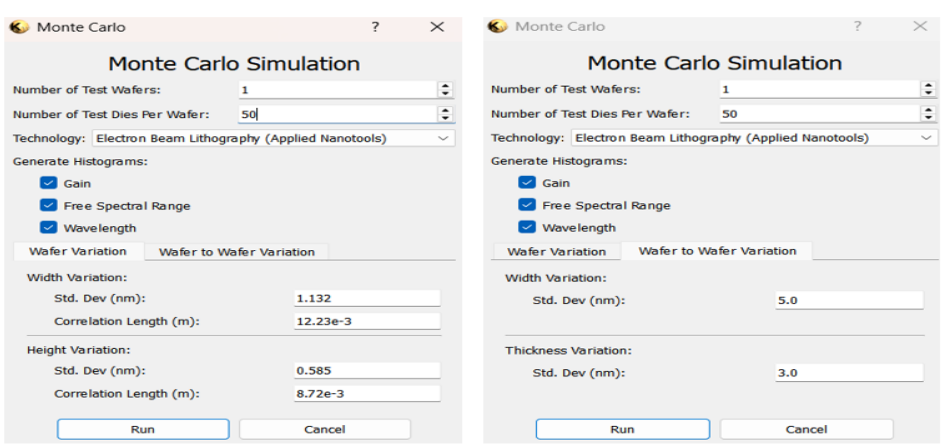

Fig. 37: Monte Carlo simulation settings

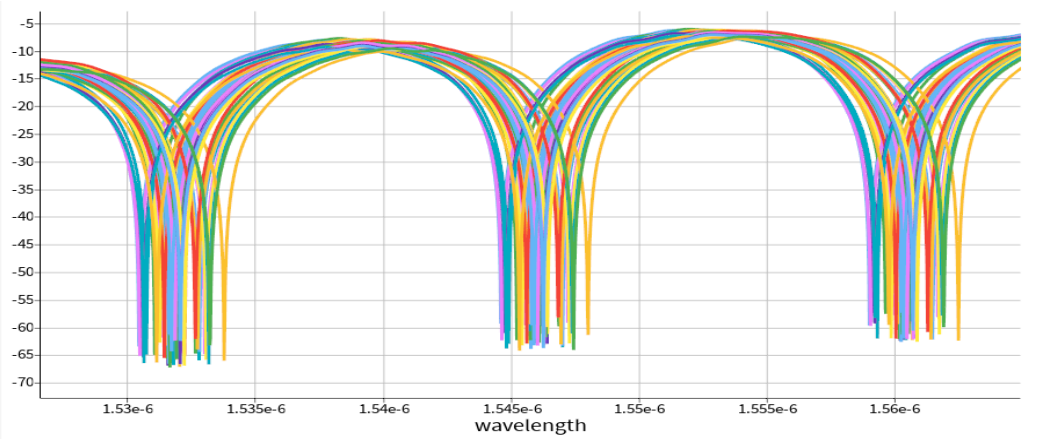

Fig. 38: Monte Carlo simulation results for the **TE** MZI

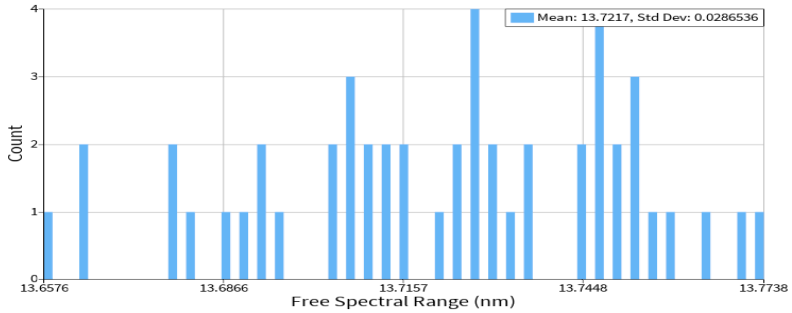

Fig. 39: Histogram of the TE MZI (ΔL = 40 µm) FSR

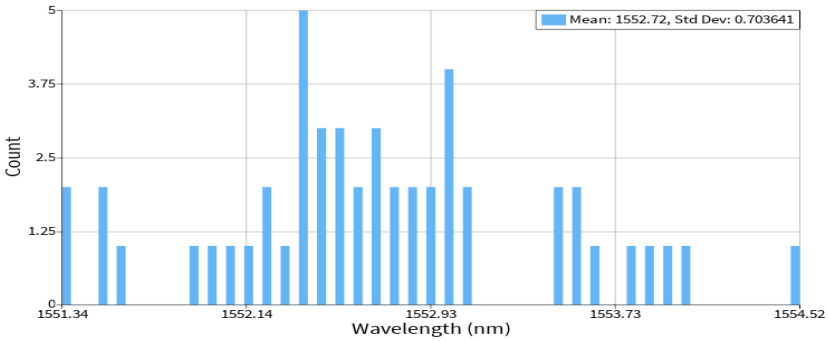

Fig. 40: Histogram of the TE MZI (ΔL = 40 µm) wavelength peak

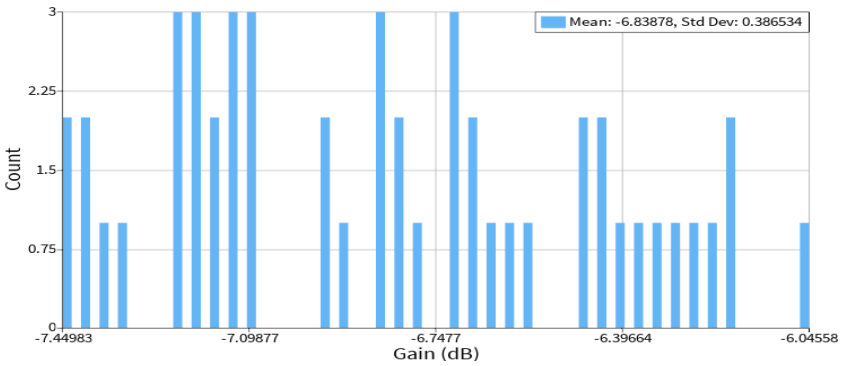

Fig. 41: Histogram of the TE MZI (ΔL = 40 µm) gain

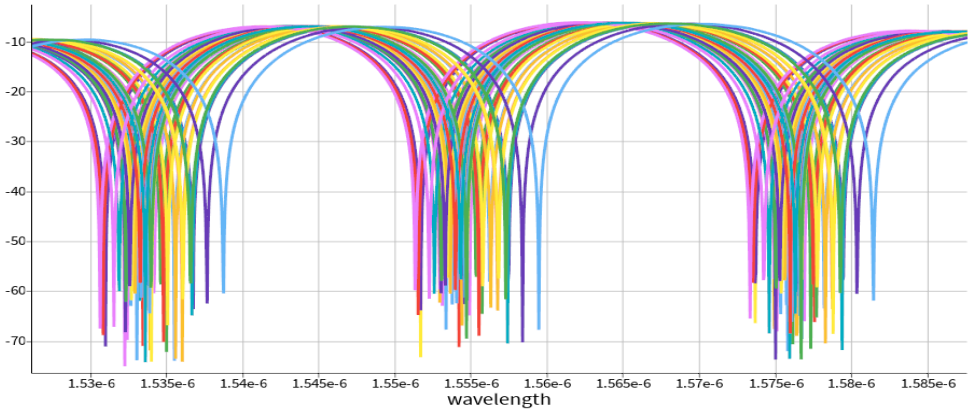

Fig. 42: Monte Carlo simulation results for the **TM** MZI

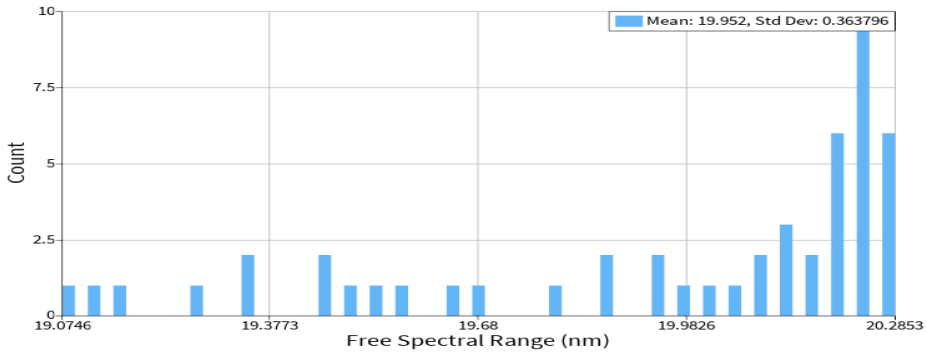

Fig. 43: Histogram of the TM MZI (ΔL = 30.035 µm) FSR

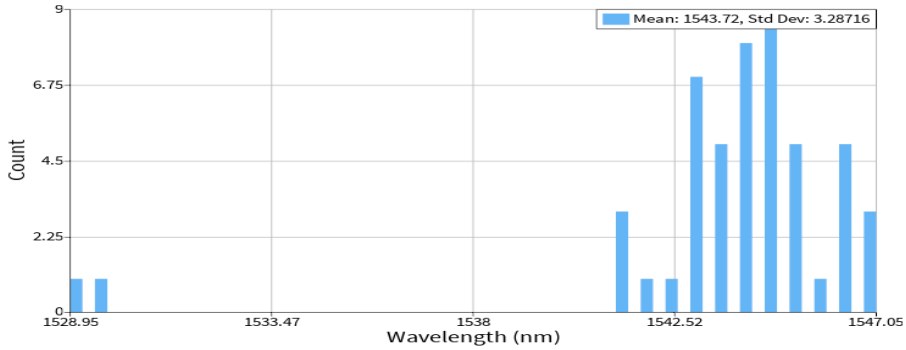

Fig. 44: Histogram of the TM MZI (ΔL = 30.035 µm) wavelength peak

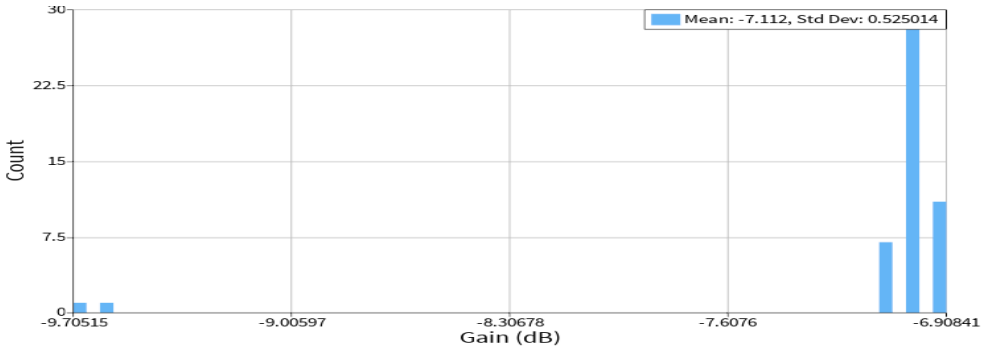

Fig. 45: Histogram of the TM MZI (ΔL = 30.035 µm) gain

6. Measurement Data Analysis

To characterize the devices, a custom-built automated test setup with automated control software written in Python was used. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers. A polarization maintaining fibre array was used to couple light in/out of the chip.
Based on the fiber array and detector configuration in the measurement setup, the optical power collected from detector1, detector2, and detector3 are labeled as channel_1, channel_2, and channel_3, respectively.

6.1. Data Analysis of the TE Mode MZI

First, we removed the baseline shape of the grating couplers. To do that, we curve-fit the spectrum using a 3-order polynomial. Then, subtract this polynomial from subsequent data to get the baseline corrected and flattened results. I used the data from the **Scylla Stage**.

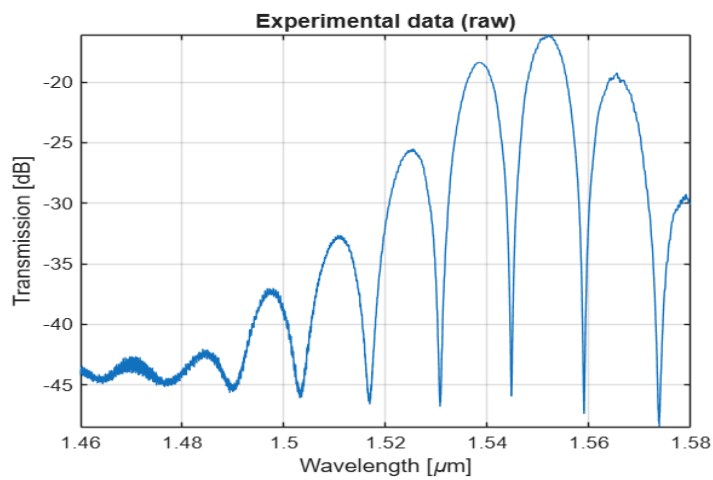

Fig. 46: Raw experimental data of the TE mode MZI

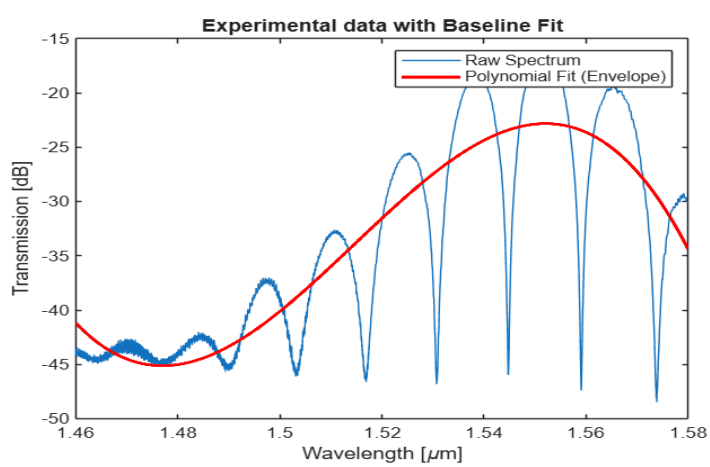

Fig. 47: Polynomial fit of the TE mode MZI

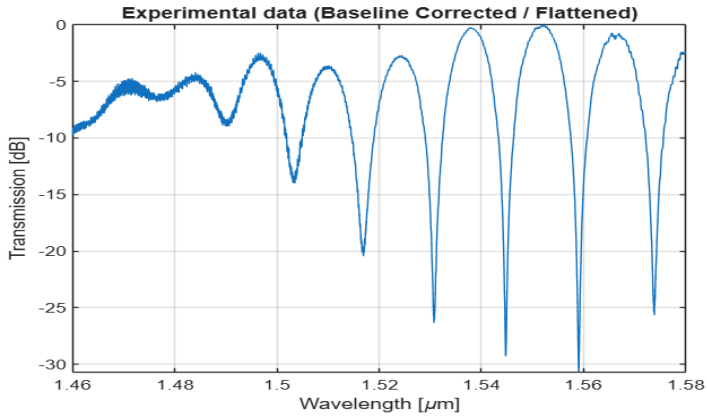

Fig. 48: Baseline Corrected result of the TE mode MZI

Loop back calibration structures are used to eliminate the effects of grating couplers from MZI transmission spectrum for both TE and TM modes. A Loop back structure consists of two grating couplers connected by a waveguide path. The grating couplers gives a non-flat spectral response which is bandwidth limited, and we want to remove this effect.

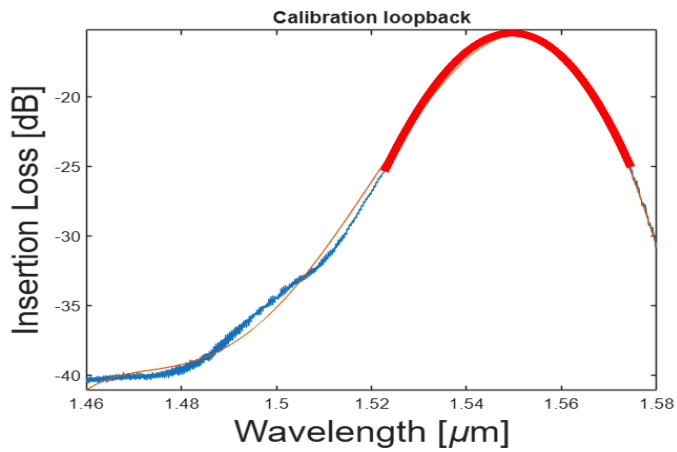

Fig. 49: Loop-back calibration for the TE mode MZI

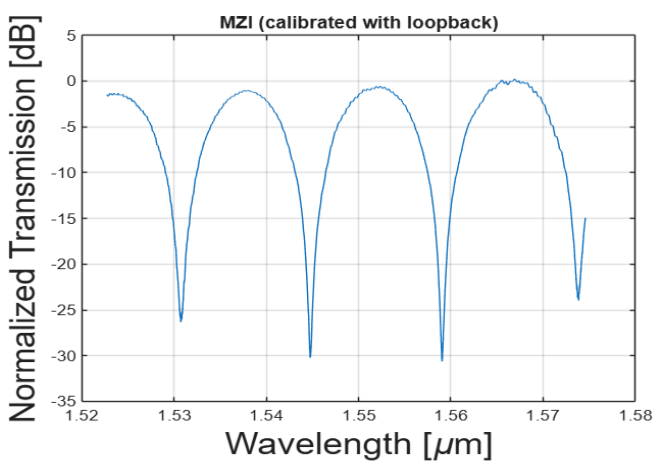

Fig. 50: Loop-back calibrated TE mode MZI

Next, we try to find the peak of the spectrum to extract the FSR and group index and also find the parameters to better fit the MZI model.

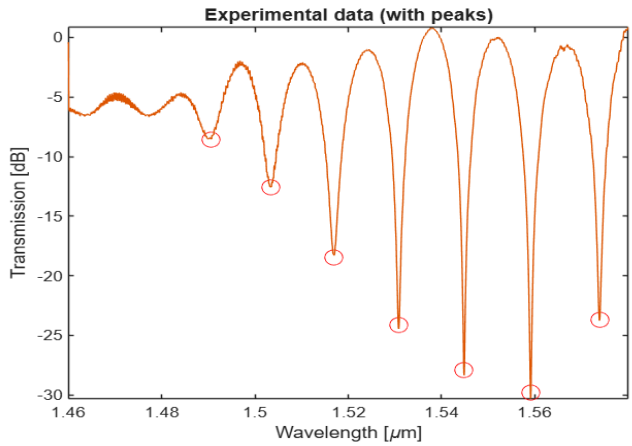

Fig. 51: Finding peaks of the TE mode MZI

(estimate) Group index: 4.2296
Dispersion [ps/nm/km]: -7530.2105

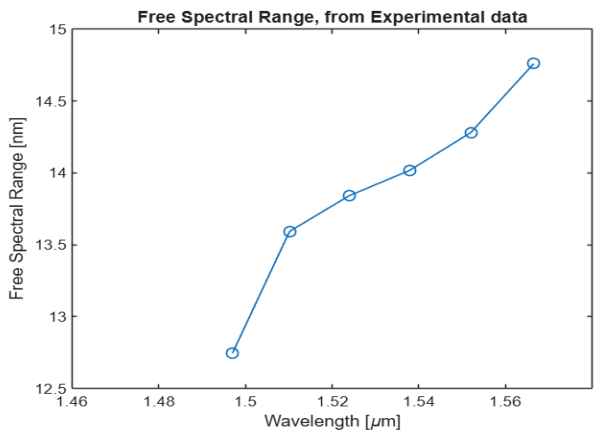

Fig. 52: Measured FSR of the TE mode MZI

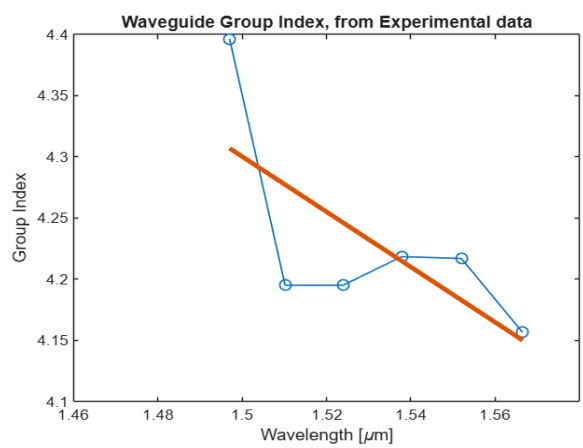

Fig. 53: Measured group index of the TE mode MZI

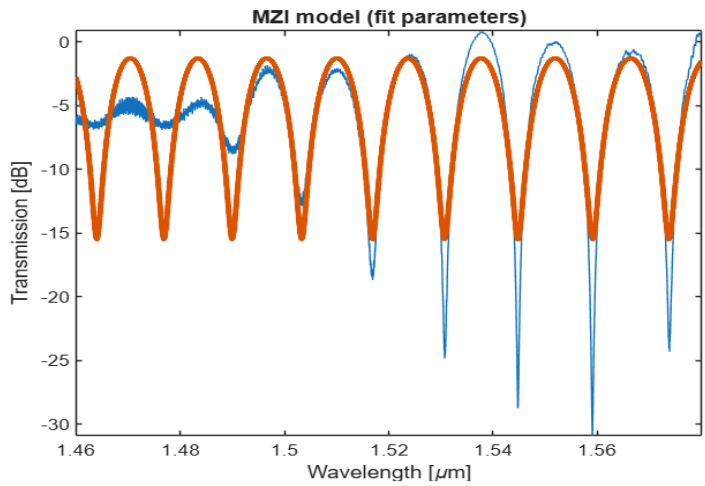

Fig. 54: TE MZI model

We can also use the autocorrelation method.

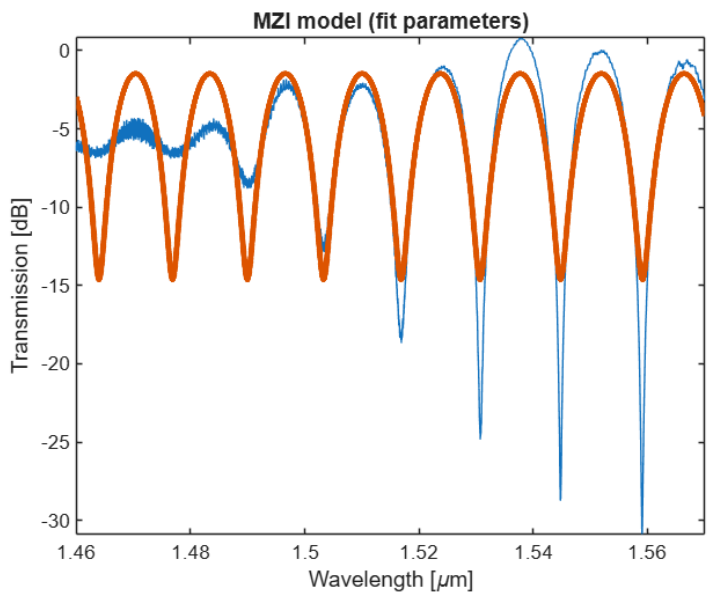

Fig. 55: TE MZI model

6.2. Data Analysis of the TM Mode MZI

Similar to the TE MZI analysis, we can repeat the same procedure for the TM MZI measurement results.

- The measurement results from the Scylla stage for the TM MZI with ΔL= 30.035 µm was missing, and the results for the TM MZI with ΔL= 12.92 µm was so noisy (Fig. 56) that the data analysis is not possible. The Iris stage didn't have the measurements of my TM devices.

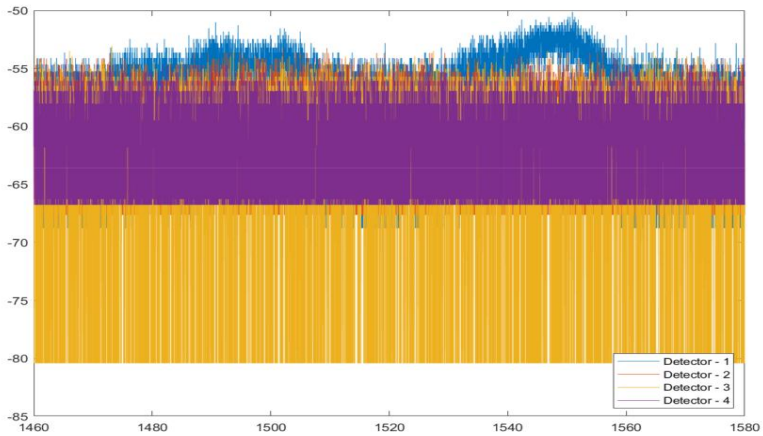

Fig. 56: Noisy measurement results of the TM MZI with ΔL= 12.92

6.2. Data Analysis of the TE Mode MRR

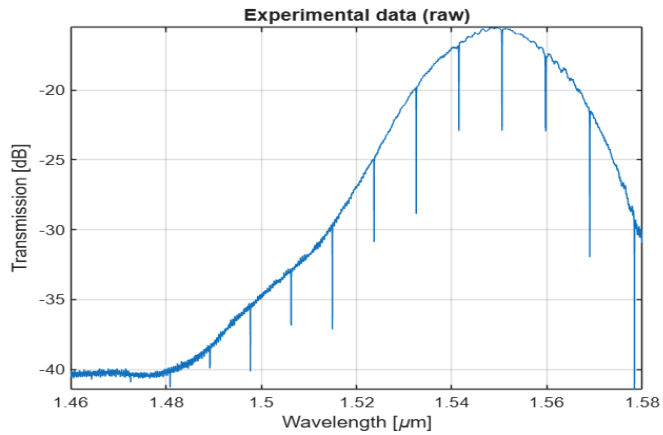

Fig. 57: Raw experimental data of the TE mode MRR

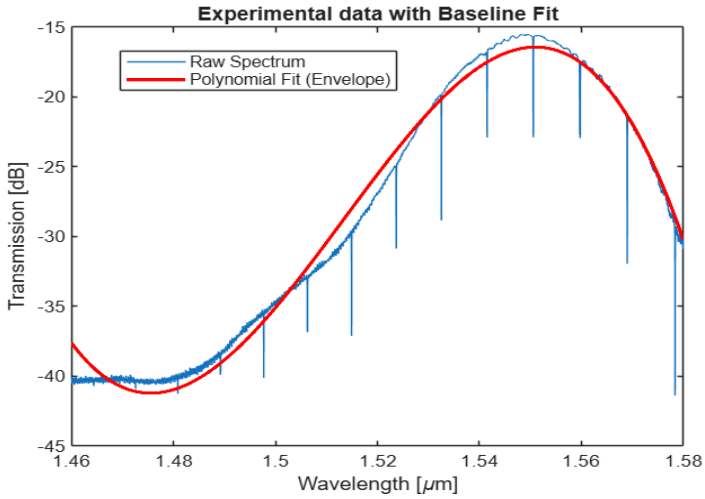

Fig. 58: Polynomial fit of the TE mode MRR

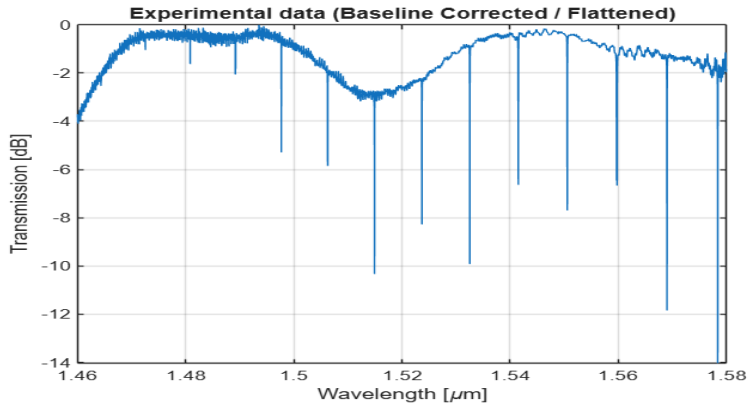

Fig. 59: Baseline corrected results of the TE mode MRR

Table 13. Comparison of the simulated and measured results for the MZI and MRR

Waveguide (nm^2)	Mode	ΔL (μm)	FSR (Sim.)	n_g (Sim.)	FSR (Exp.)	n_g (Exp.)
500 × 220	TE MZI	40	14.22	4.2238	14.28	4.2296
500 × 220	TM MZI	30.035	21.31	3.7536	-	-
500 × 220	TE MRR	L=2πR	8.61	4.44	9.2	4.156

The measurement results fit within the corners in Table 12.

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